SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	Sameer Ahmed G	**Reg. No.:**	192421154
Section / Batch:	B-TECH IT	**Date:**	28/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct (192421154)

I Sameer Ahmed G (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Ahy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

Closest Pair - Weather Station Proximity Analysis

Given points:

A(1,1), B(3,4), C(6,6), D(2,2), E(8,1), F(5,5)

Distance Formula:

$$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

Distance Calculations:

A(1,1)

$AB = \sqrt{(3-1)^2 + (4-1)^2} = \sqrt{4+9} = \sqrt{13} = 3.61$

$AC = \sqrt{(6-1)^2 + (6-1)^2} = \sqrt{25+25} = \sqrt{50} = 7.07$

$AD = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

$AE = \sqrt{(8-1)^2 + (1-1)^2} = \sqrt{16+16} = \sqrt{32} = 5.66$

$AF = \sqrt{(5-1)^2 + (5-1)^2} = \sqrt{49} = 7$

B(3,4)

$BC = \sqrt{(6-3)^2 + (6-4)^2} = \sqrt{9+4} = \sqrt{13} = 3.61$

$BD = \sqrt{(2-3)^2 + (2-4)^2} = \sqrt{1+4} = \sqrt{5} = 2.24$

$BE = \sqrt{(8-3)^2 + (1-4)^2} = \sqrt{25+9} = \sqrt{34} = 5.83$

$BF = \sqrt{(5-3)^2 + (5-4)^2} = \sqrt{4+1} = \sqrt{5} = 2.24$

C(6,6)

$CD = \sqrt{(2-6)^2 + (2-6)^2} = \sqrt{16+16} = \sqrt{32} = 5.66$

$CE = \sqrt{29} = 5.39$

$CF = \sqrt{2} = 1.41$

D(2,2)

$(D,E) = \sqrt{(2-8)^2+(2-1)^2} = \sqrt{37} = 6.08$

$DF = \sqrt{(2-5)^2+(2-5)^2} = \sqrt{18} = 4.24$

E(8,1)

$EF = \sqrt{(8-5)^2+(1-5)^2} = \sqrt{25} = 5$

Closest Pair:

The minimum distance obtained is

$\sqrt{2} = 1.41$

Therefore the closest weather station pairs are:

- A(1,1) and D(2,2)
- C(6,6) and F(5,5)

Number of Comparisons:

$$\frac{n(n-1)}{2}$$

For $n=6$:

$$\frac{6 \times 5}{2} = 15$$

$\therefore$ Total Comparisons $= 15$

Thus the Brute Force closest pair algorithm has quadratic time complexity $O(n^2)$.